



INFORMATION TECHNOLOGY

gration of digital technologies to provide more efficient urban services will help to facilitate fine communication between the government and citizens.

Redefined role of data centres

Development of smart cities requires a serious rethink of how the infrastructure is designed and deployed. It focuses on initiatives that increase capacity and energy efficiency at the same time. Inclusion of technologies like the IoT, Big Data and AI has generated a lot of excitement about the possibilities of smart cities.

Smart cities will generate vast amounts of data from all connected devices, which will keep growing exponentially. So, it is a fundamental requirement to set up the technical infrastructure in place to store this valuable data. It is equally important to have an abundant and reliable power supply to keep the whole infrastructure working.

Data centres will be the major contributors for the emergence of smart cities, as these can address such crucial requirements as ample connectivity,

huge data storage and high compute power for analytics to crunch all data.

The eco-friendly factor drives the way

One of the main objectives of smart cities is to address the current environmental challenges. Smart cities promise to improve energy efficiency by bringing in advanced applications that track individual and corporate emissions, and to suggest ways to achieve this. Moreover, organisations need to meet the environment-friendly standards and set up eco-friendly IT infrastructure to comply with the government's Smart Cities initiative.

Modular data centres have an edge in reducing organisational power consumption. Their power usage effectiveness (PUE) figures are expected to be much lower than those of traditional data centres. However, considering the fact that there is always a need for larger, core data centres to deliver organisations' digital services, data centre optimisation processes can be opted for by retuning these with smarter cooling

solutions. This can largely increase energy efficiency and IT capacity of organisations, enabling them to take advantage of the emerging IT landscape.

Shifting to co-located facilities also plays a key role in contributing to the eco-friendly parameters of the smart cities. Various SMBs and large-sized enterprises that need to expand their infrastructure capacity should look towards hosting co-located data centres, which offer secure and safe access to data. This is also a simple way to improve energy efficiency of organisations.

Data centres contribute to smart cities: an undeniable fact

Data centres bridge the gap between edge and core levels of smart city architecture. Advanced monitoring of data flow and cutting-edge management technologies allow the new data centre environment to be controlled from a single glass pane. Futurists and experts say that data centres are one of the main building blocks to bring smart cities to reality. **EFY**



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